

ORIGINAL ARTICLE

Species-specific functional traits rather than phylogenetic relatedness better predict future range-shift responses of odonates

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Funding information

Agència de Gestió d'Ajuts Universitaris i de Recerca, Grant/Award Number: BP-DGR-2011

Editor/associate editor: Christopher Hassall

Abstract

1. Climate change has the potential to modify habitat characteristics and, consequently, induce species responses to ongoing environmental changes. Functional traits determine both if a species can persist and maintain stable population sizes in particular ecological conditions, and its capacity to disperse to more favourable habitats.
2. Given that functional traits evolve over time, one could expect closely related species to show similar responses to climate change, which should identify vulnerable lineages. Alternatively, species-specific functional traits may anticipate species responses to climate change, and therefore, trait composition should be a strong predictor.
3. We compiled a comprehensive dataset of functional traits of 84 Iberian and Moroccan odonates species and built a phylogenetic tree to determine if dissimilarity of traits and phylogenetic relatedness are relevant to better discriminate species range-shift responses to climate change.
4. Modelling results for 66 species showed clear impacts of the increase in temperature and drought events on their potential distribution. The traits that best-discriminated species that expanded their ranges were multivoltinism, short-life cycles and preference for temporary habitats, whereas species with a reduced and displaced potential distribution were mostly semivoltine, with a short flying season, oviposition on gravel and restricted to permanent streams, small rivers or

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oligotrophic lakes. Trait conservatism was rejected and phylogenetic relatedness was a poor predictor of range shifts.

5. Considering odonates as model organisms, traits such as voltinism, beginning and prolongation of flight period and preference of temporal habitats should be examined to anticipate range-shift responses of freshwater insects to climate change.

KEYWORDS

dispersal, dragonflies, evolutionary history, extinction, functional traits, global change, potential distribution

INTRODUCTION

Predicted rates of climate warming will modify current habitat characteristics for many species (Chen et al., 2011), prompting responses that will likely differ across taxa (Buckley & Kingsolver, 2012). Many species are expected to expand or shift their geographical ranges to higher latitudes and/or elevations, where ecological conditions may match their current environmental niche and where they may be able to establish viable populations (Angert et al., 2011; Markovic et al., 2014). Other species may persist in situ through local adaptation or phenotypic acclimation (Buckley & Kingsolver, 2012; Stoks et al., 2014). Yet other species, unable to cope with the new ecological conditions, will suffer population declines (Angert et al., 2011; Hassall & Thompson, 2008) and may even go extinct (Parmesan, 2006). Understanding which environmental drivers and functional traits most strongly influence species responses to climate change should increase our ability to assess species vulnerability, defined as 'the predisposition to be adversely affected' (Oppenheimer et al., 2015).

The set of biotic and abiotic conditions under which a given species can persist and maintain stable population sizes is determined by its functional traits (Violle et al., 2007). Therefore, functional traits composition should modulate responses to climate change and determine species vulnerability. For instance, generalist species with high phenotypic plasticity and capable of using available resources will have the ability to adapt in response to environmental changes (Parmesan, 2006; Stoks et al., 2014). In contrast, species with a strong phenotype-environment match, for instance, ecological specialists requiring particular environmental conditions with narrow physiological tolerances or high dependence on specific resources, will be sensitive to climate change and their capacity to find favourable habitats will depend on dispersal abilities (Buckley & Kingsolver, 2012). Therefore, elucidating which traits modulate species responses to climate change can help to anticipate distribution shifts and ultimately to identify vulnerable species for improving conservation strategies (Estrada et al., 2016; Pacifici et al., 2015).

Species functional traits evolve over time in response to species interactions and changing ecological conditions. Despite the convergence of some traits between lineages, trait evolution is constrained by ancestral ecological characteristics and lineages may retain large portions of their functional trait composition through time (Múrria et al., 2018; Múrria et al., 2020), which is known as phylogenetic trait conservatism (Webb et al., 2002). Accordingly, one can expect closely

related species to show similar responses to climate change because of overlapping functional traits composition (Buckley & Kingsolver, 2012). Thus, if functional traits associated to a potential response to climate change (i.e., oviposition habitat or duration of life cycle) were phylogenetically preserved, closely related species would likely show comparable responses to climate change (Buckley & Kingsolver, 2012). If this was the case, phylogenetic relatedness would help anticipate patterns of future responses to climate change (Fei et al., 2017).

Aquatic ecosystems have been shown to be highly vulnerable to global change due to habitat loss, decreased habitat heterogeneity, reduced connectivity and additional stressors such as pollution, river regulation, over-abstraction of water and alien species introductions (Albert et al., 2021; Woodward et al., 2010). Moreover, the expected increase in temperature and in the frequency and duration of drought events will influence physiological processes of freshwater macroinvertebrate species, including adjustments in life cycle duration and reductions in body size (Burgmer et al., 2007; Stoks et al., 2014). As a result, suitable habitat ranges will likely change (either expanding or contracting) for many species and the composition of freshwater macroinvertebrate communities will probably be modified (Domisch et al., 2013; Markovic et al., 2014). Clear predictions of future habitat alteration and associated species responses to these changes are critical for improving conservation and management of freshwater ecosystems and their biodiversity. However, it is still unclear whether functional traits can predict differences in the extent of range shifts in response to climate change (Angert et al., 2011; Rocha-Ortega et al., 2020).

Among freshwater macroinvertebrates, Odonata (Insecta) are a good model taxon for exploring the effects of climate warming on freshwater biodiversity because (1) they originated in the tropics, which implies tolerance to high temperatures; (2) adults present high morphological differentiation across species, which facilitates capture, identification and recording; (3) they occur with medium to high local abundances and are present in almost all freshwater habitats, with high species turnover due to larval and adult preferences in habitat use and evident niche partition among species; (4) they present high phenotypic variability, both within and between species, in traits presumably involved in adaptation to climate warming, such as emergence and duration of flight season; and (5) there is a large and well-known body of historical research, including extensive recording and rich historical datasets from citizen science projects (Corser et al., 2014; Hassall & Thompson, 2008; Termaat et al., 2019).

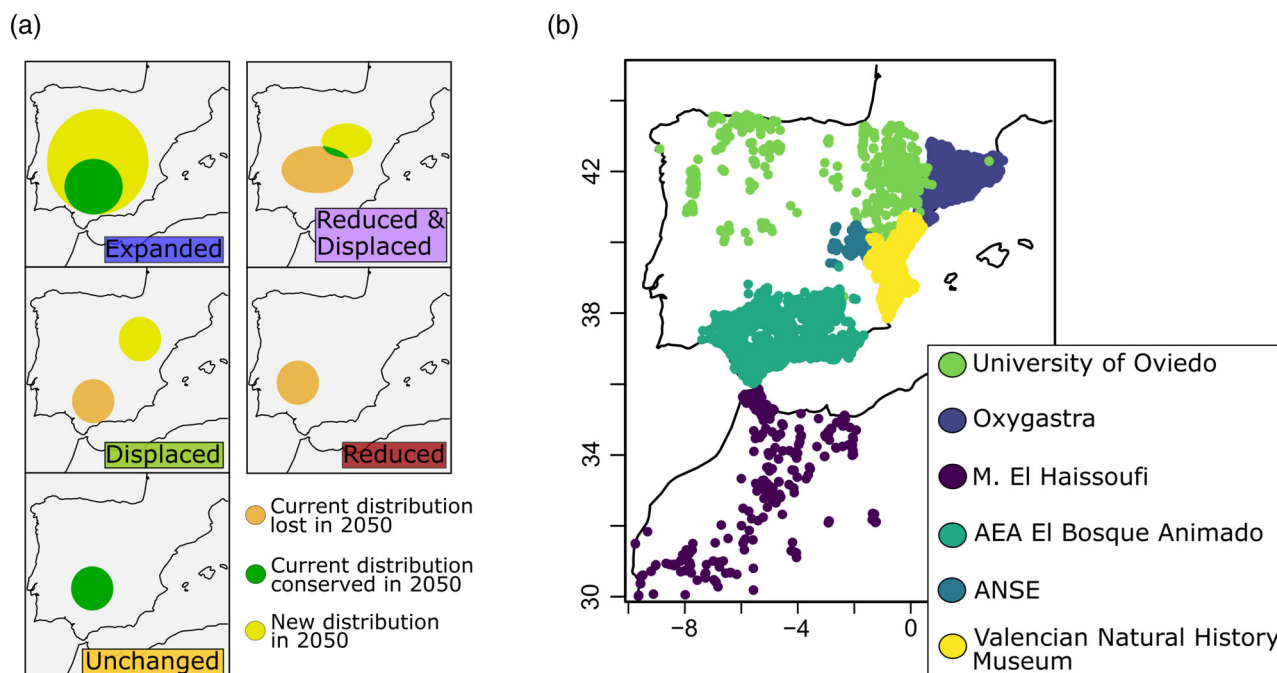


FIGURE 1 (a) Proposed categories of potential range shifts based on differences between current and future species distribution in terms of longitude and latitude; (b) species occurrence map covering the Iberian Peninsula and Northern Morocco from six datasets.

The thermal sensitivity of odonates drives species distribution and turnover along the latitudinal gradient and makes them a potential indicator for environmental change (Dingemanse & Kalkman, 2008; Hassall et al., 2007; Rocha-Ortega et al., 2020). Concurrently with climate warming in Central and Northern Europe, a clear trend of rapid northward range expansion of odonates has been observed (Markovic et al., 2014; Termaat et al., 2019). For instance, at least 15 Mediterranean species have expanded their geographic range to Central and Northern Europe since 2009, whereas some African species have expanded to Southern Europe (Ott, 2010). In the Iberian Peninsula, at least six African species have arrived from Morocco during the last 60 years and are currently experiencing northward expansion. Climate warming may also be changing the phenology of odonates; for instance, an earlier beginning and a prolongation of the flight season have been recorded (Dingemanse & Kalkman, 2008; Richter et al., 2008).

Across the Iberian Peninsula, we expect odonates to show different responses to climate change. For instance, taxa that prefer temporary habitats such as ponds or intermittent streams, which present fast development rates and show good dispersal abilities, should expand their range. In contrast, lineages with poor dispersal abilities or living at high elevations should reduce or shift their distributions, respectively (Grewe et al., 2013). As a result, we propose five range-shift categories associated to climate change based on the differences on range and overlapping between the modelled current and future potential distributions of suitable habitat of each species (Figure 1a): 'Expanded', 'Displaced', 'Unchanged', 'Reduced and displaced' and 'Reduced'.

Here, using a database of 84 Iberian and Moroccan (Westernmost Mediterranean region) Odonata species, we explore how functional traits and phylogenetic relatedness can fit the a priori-defined

categories of range shifts. First, we aim to assess if the dissimilarity of functional traits can discriminate species clustered in different categories of range-shift response and to elucidate which traits are representative for (i.e., better discriminate) a given category. Second, we aim to explore if phylogenetic relatedness corresponds to categories of range-shift responses, which would indicate that phylogenetic trait conservatism is relevant for anticipating future species responses (Davis et al., 2010). In either case (if closely related species were placed in distinct range-shift categories, or if acquired species-specific functional traits better-predicted species range shifts), we could profit from a tool to anticipate species responses to climate change and guide conservation efforts for odonates. The alternative, that is, idiosyncratic responses, would imply the need for species-specific management plans.

MATERIALS AND METHODS

Species records

The study area covered the westernmost Mediterranean region with a total number of 84 species, 25 occurring exclusively in the Iberian Peninsula and 9 restricted to Morocco (Torralba-Burrial & Ocharan, 2013). Occurrence of adults was compiled from six datasets (Figure 1b, see details at 'Data Availability Statement'): (1) Oxygastra, the Catalan Odonata Study Group; (2) ROLA's project of the environmental education association 'El Bosque Animado'; (3) the Arthropod Collection of the University of Oviedo (Torralba-Burrial & Ocharan, 2013); (4) the Valencian Natural History Museum; (5) the

Asociación de Naturalistas del Sureste (ANSE); and (6) the Moroccan dataset collected by Mohamed El Haissoufi. When combining datasets, only records with taxonomic identification at the species level and observations after 2003, when more data and precise spatial coordinates started to be recorded systematically, were considered. For modelling purposes, in order to homogenise the sample effort and to reduce the spatial autocorrelation associated with environmental variables (see below), the presences of a species in a 1-km² grid cell were brought together and defined as a single detection, whereas the non-occurrence of a species at a given 1-km² cell over several years of sampling was considered as a true absence (Supporting information S1). The data used are currently the main contribution of the Iberian data at gbif.org and at the Atlas of the European dragonflies and damselflies (Boudot & Kalkman, 2015). However, we avoided using data records directly from GBIF because they were collected following different protocols in each region and also including 'opportunistic' species without a harmonised sampling effort or geographical representativeness.

Climatic models and current and future predictions of potential distributions

In order to assess how individual species potential distributions could change in the future based on current occurrences and using climatic conditions as predictors, species distribution models (SDMs) were used. SDM methods have been widely debated because future predictions are frequently over-interpreted (Araújo & Peterson, 2012), especially when limited occurrence data in high-dimensional environmental space are used (Peterson, 2011). While acknowledging this criticism, our intent was not to accurately map the future ranges of species, but to evaluate the models and classify the species into the five proposed range-shift categories. Since model performance decreases with sample size (Stockwell & Peterson, 2002), 10 species were discarded because they had been recorded in less than 20 localities (Supporting information S1).

The geographic extent of bioclimatic variables to model current distribution covered northern Morocco, the Iberian Peninsula and southern France, between latitudes 30° N and 45° N, and longitudes 10° W and 5° E. The predictions of future potential distribution were exclusively limited to the Iberian Peninsula and South of France. Current climatic conditions were described by 19 bioclimatic variables based on temperature and rainfall values. These variables were downloaded from worldclim.org at 30 arc-second resolution based on monthly averages for the period 1970–2000. Bioclimatic variables are commonly used in SDMs for predicting species ranges and were originally created to generate more biologically meaningful variables than traditional environmental metrics such as mean temperature and precipitation (O'Donnell & Ignizio, 2012).

To remove the redundant information of bioclimatic variables, Spearman correlation and variance inflation factor (VIF) analyses were performed with all bioclimatic values extracted from all occurrence points. To purge highly correlated bioclimatic variables, the biological and ecological meaning of each variable was considered. Finally, the

five less correlated bioclimatic variables were selected: Isothermality (*Bio 3*), which quantifies the monthly mean diurnal range relative to the annual temperature oscillation; temperature seasonality (*Bio 4*), which is a measure of annual temperature variability; Mean Temperature of Wettest Quarter (*Bio 8*), which is calculated as the average temperature of the three consecutive wettest months; Mean Temperature of Driest Quarter (*Bio 9*), which is the average temperature of the three consecutive driest months; and Precipitation Seasonality (*Bio 15*), which is a measure of monthly total precipitation variability. Final models were produced for each species after variable selection (with linear and quadratic terms) by their mean values (Supporting information S2).

For predicting species potential distributions in 2050, the HadGEM2-ES climate model (Met Office Hadley Centre, MOHC, and Instituto Nacional de Pesquisas Espaciais) was selected because it focuses on land components (Bellouin et al., 2011). To capture the potential effects of climate change on species distributions, the lowest and the highest anthropogenic emissions scenarios were considered (RCP 2.6 and RCP 8.5, respectively). Predicted values for 2050 for the five selected bioclimatic variables were downloaded from ccafs-climate.org in ASCII format, again at 30 arc-second resolution.

To determine the current species distribution, binary logistic generalised linear models (GLM) and boosted regression trees (BRT) were run for each species because testing for several algorithms is suggested to optimise the predictions for each particular species (Konowalik & Nosol, 2021; Qiao et al., 2015). GLM represents a flexible extension of linear models that allows for response variables that have a non-normal distribution error (binomial in the case of binary responses), whereas BRT combines regression trees for generating large numbers of simple distribution models that are selected for the best prediction by boosting algorithms (Elith et al., 2008; Franklin, 2010; Kienast et al., 2012). To create the input SDM matrix, the current bioclimatic information was extracted for each spatial point. The database created for each species was split into 70% as training set and 30% as testing set by random partition to validate the models. Each model was evaluated independently by three predictors of significance to select the model with the best performance: (1) by means of area under curve (AUC) statistics from a receiver-operating characteristic analysis (ROC), which is a threshold-independent evaluation of model discrimination (Fielding & Bell, 1997); (2) by true skills statistics (TSS), which is a measure of accuracy (Allouche et al., 2006) and (3) by odds ratio skill score (ORSS), which measures skill compared to a random prediction (Stephenson, 2000). AUC and TSS were the two metrics that achieved the highest performance compared to other evaluation models (Konowalik & Nosol, 2021) (Supporting information S2). The widely used logistic regression (fitted with GLM) was selected for the majority of species. However, for some species that had low numbers of occurrences (fewer than 70 grid cells), the BRT models (learning rate = 0.001, bag fraction = 0.75) performed better than GLM, as previously suggested (Konowalik & Nosol, 2021). All models were performed and evaluated using the R software (R Core Team, 2016) and the package 'gbm' (Ridgeway et al., 2015).

The probability distribution maps of current and future projections were transformed into binary presence-absence maps with a

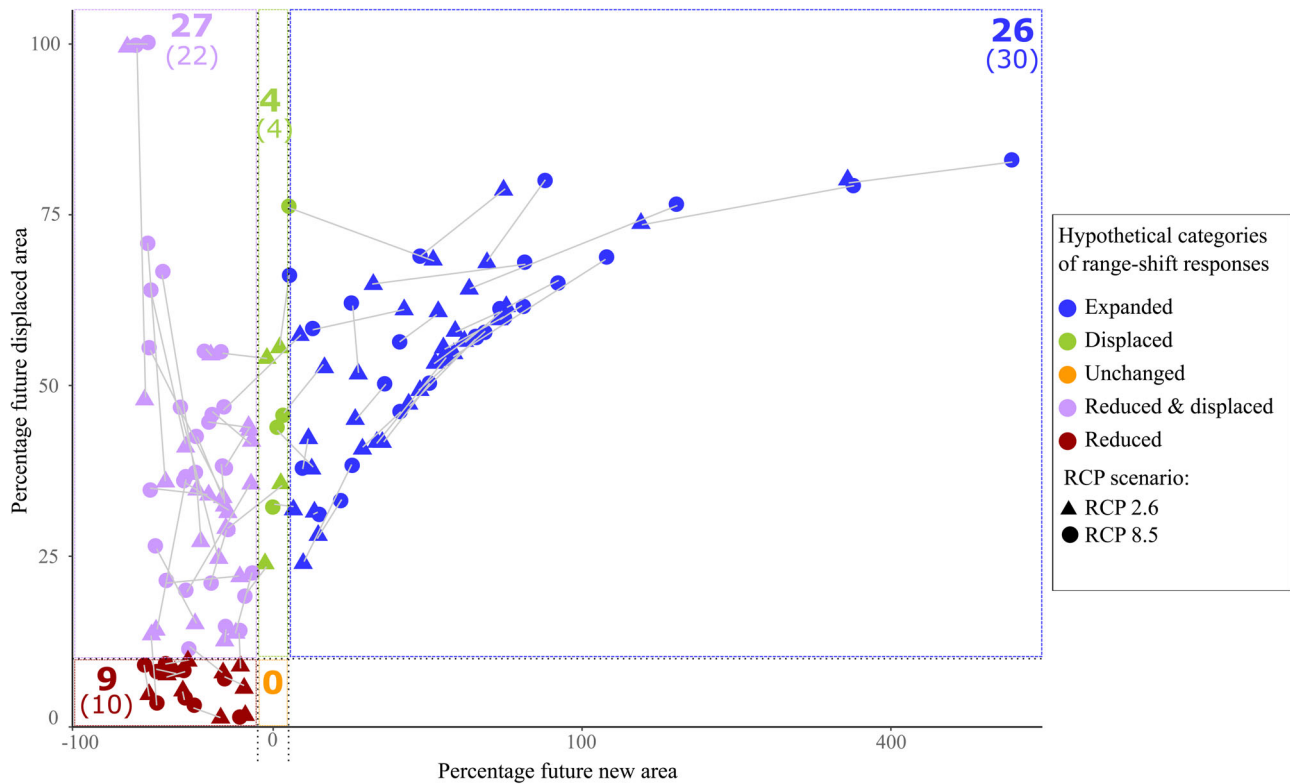


FIGURE 2 Classification of species into categories of range-shift responses considering the high emissions RCP 8.5 model (circles and number of species in bold) and the low emissions RCP 2.6 model (triangles and number of species in parentheses). Lines connect the positions for both emission scenarios.

model-specific threshold obtained as the cut-off value that minimised the difference between sensitivity (true positive rate) and specificity (true negative rate) (Lobo et al., 2007) using the R package ‘dismo’ (Hijmans et al., 2016). Differences between current and future predictions of SDM for each species were used to assign species into the five proposed categories of range shifts. This is shown in Figure 2, where the x-axis represents the percentage increase of occupied area, calculated as:

$$\text{Formula1(AxisX)} := \frac{\sum \text{futurepixels} - \sum \text{currentpixels}}{\sum \text{currentpixels}} \cdot 100$$

The y-axis represents the percentage of displaced area (i.e., pixels predicted to be occupied in the future but not currently occupied) calculated as:

$$\text{Formula2(AxisY)} := \frac{\sum \text{newlyoccupiedpixels}}{\sum \text{futurepixels}} \cdot 100$$

Within this space, the proposed categories of range shifts were operationally defined as: *Expanded*: Axis X $\in [10, +\infty)$, axis Y $\in [10, +\infty)$; *Displaced*: Axis X $\in (-10, +10)$, axis Y ≥ 10 ; *Unchanged*: Axis X $\in (-10, +10)$, axis Y $\in [0, 10]$; *Reduced and displaced*: Axis X < -10 , axis Y ≥ 10 ; and *Reduced*: Axis X ≤ -10 , axis Y $\in [0, 10]$.

Species-specific functional traits

Information on functional traits for each species was extracted mainly from D’Aguilar et al. (1986), Dijkstra (2006), Corbet (2004), Corbet et al. (2006), Hassall and Thompson (2008), Powney et al. (2014), Boudot and Kalkman (2015) and Martín et al. (2016), plus specific literature (e.g., Rodríguez-Martínez & Torralba-Burrial, 2012; Samraoui et al., 1998). We focused on traits associated to site fidelity, dispersal ability, reproductive strategies, competitive abilities and persistence in unfavourable climatic conditions (Estrada et al., 2016). Traits were quantified using a fuzzy coding approach (Chevenet et al., 1994) (Supporting information S3). The final matrix included 15 traits and 49 categories (Supporting information S4): one morphological trait (body length), three distributional characteristics (local abundance, distribution width and range margins), four traits associated to habitat preferences (habitat type and seasonality, vegetation, water quality) and seven biological traits (flight season, flight behaviour, dispersal behaviour, voltinism, drought resistance, oviposition habitat and egg diapause). All scores within each trait and species were standardised so that they added up to 1. To visualise the ecological space of species based on functional trait composition in reduced dimension we conducted fuzzy principal component analysis (fPCA, Chevenet et al., 1994) using the *dudi.fPCA* function of the R package ‘ade4’ (Dray & Dufour, 2007). The variable contribution of each axis was

calculated as the correlation between the original trait categories and the functional space axes.

Compilation of DNA sequences and phylogenetic analyses

DNA sequences for all available genes of all Iberian and Moroccan species were compiled from the GenBank database. The final list of genes included the mitochondrial cytochrome c oxidase subunit I gene (COI; 587 bp) and the ribosomal 12S rRNA (12S rRNA; 1774 bp), 16S rRNA (16S rRNA; 542 bp), 18S rRNA (18S rRNA; 1813 bp) and 28S rRNA (28S rRNA; 3933 bp). As outgroups, the four Ephemeroptera (Insecta) species *Baetis harrisoni*, *Callibaetis ferrugineus*, *Ephemera danica* and *E. orientalis* were used (Supporting information S5). Methods to generate the phylogenetic tree are described in Supplementary Information S6.

Data analysis

To assess whether selected functional traits could be used to classify a species into one of the range-shift categories, we applied discriminant analysis on fuzzy principal components using the *discrimin* function in the R package 'ade4' (Chessel et al., 2004). To select the subset of traits that significantly contributed to discriminating among categories, we implemented a backwards selection procedure in R using a simple pseudo-*F* ratio as a measure of discriminating power, and a permutation test to obtain a *p*-value. The pseudo-*F*, modelled after Fisher's *F* statistic, was calculated as the between-groups mean sums of the squared distances among points (i.e., species) in the discriminant analysis space divided by the within-groups mean sums of squared distances, where distances were Euclidean distances calculated from canonical scores. In other words, a set of traits had discriminating power and the pseudo-*F* was high if the groups of species defined by categories were apart and compact in the discriminant analysis space.

Given this statistic, the stepwise backwards selection proceeded as follows. We started with a full discriminant analysis model that included all traits and redid the analysis with all traits minus one, removing traits one at a time. For each of the traits that were removed, a new pseudo-*F* was calculated, and traits were ordered by the reduction in the value of this statistic with respect to the full model. If the reduction in pseudo-*F* was relatively large after removing a particular trait, that trait could be thought of as contributing substantially to discriminating among categories, and vice versa. The trait which produced the smallest reduction in pseudo-*F* after removal was eliminated unless that reduction was statistically significant. To evaluate the significance and obtain a *p*-value for a given trait, the discriminant analysis with the full model was redone 999 times after randomising the observed fuzzy values for that trait across species (and therefore across categories). A *p*-value was obtained as the fraction of values (the observed pseudo-*F* value plus the 999 values from

permutations) that were equal to or greater than the observed pseudo-*F* (because if a trait is useful for discrimination, a decrease in pseudo-*F* after permuting that trait should be observed). After eliminating the trait with the least discriminating power, the procedure was repeated until all traits that were left in the discriminant analysis were significant. We note that this variable selection procedure produces a parsimonious model, but does not imply that traits not selected in the final model carry no discriminating power: all the usual caveats about automatic variable selection apply here too. We run twice the analyses to compare the robustness of the results, once using all species and again with the subset of species for which the models performed well (AUC > 0.6). The results were consistent (Supporting information S2), and only results including all species are discussed.

To explore the support of character evolution and trait conservatism and the coherence between phylogenetic relatedness and defined range-shift categories, the species scores on the first and second axes on the fPCA and the assigned range-shift categories were tested on the phylogenetic tree. Two indices were calculated to infer the phylogenetic signal: Pagel's λ parameter (Freckleton et al., 2002; Pagel, 1999) and Blomberg's *K*-statistic (Blomberg et al., 2003). Pagel's λ uses likelihood ratio tests against simpler models to compare branches of the tree topology and its value ranges between 0 (absence of trait conservatism) and 1 (strong correlation between trait and the phylogenetic tree). Blomberg's *K* makes randomizations of the original trait data under a null Brownian motion model to compare pairs of branches independently of branch length. Blomberg's *K*-statistic values around 1 indicate that trait disparification follows the topology of the tree (i.e., trait conservatism), whereas a little value of *K* corresponds to a random pattern of evolution. These analyses were carried out using the R packages 'geiger' (Harmon et al., 2008) and 'picante' (Kembel et al., 2010).

RESULTS

The final dataset included 84 Odonata species and 75,767 individual recorded presences. In general, the selected bioclimatic variables showed an increase in both average temperature and temperature seasonality and a decrease in mean temperature of the wettest quarter and total precipitation and precipitation seasonality by 2050. SDMs were run for 66 out of 84 species analysed. Besides the 10 species initially removed, another four species were discarded because were rare and locally distributed, for example, *Cordulegaster bidentata* was recorded in only 30 nearby grid cells. The selected SDM was logistic regression for 60 species, whereas BRT was selected for 7 species (Supporting information S2). Of the five selected variables, the most used variables for modelling distribution were *Bio 4* (Temperature seasonality) and *Bio 15* (Precipitation seasonality), while the less utilised variable was *Bio 3* (Isothermality). The model evaluation showed high correlation ($R^2 > 0.90$) among the indices used (Supporting information S2). In general, all values of AUC were higher than 0.6 and commonly higher than 0.7, which means all predictive models are robust and reliable.

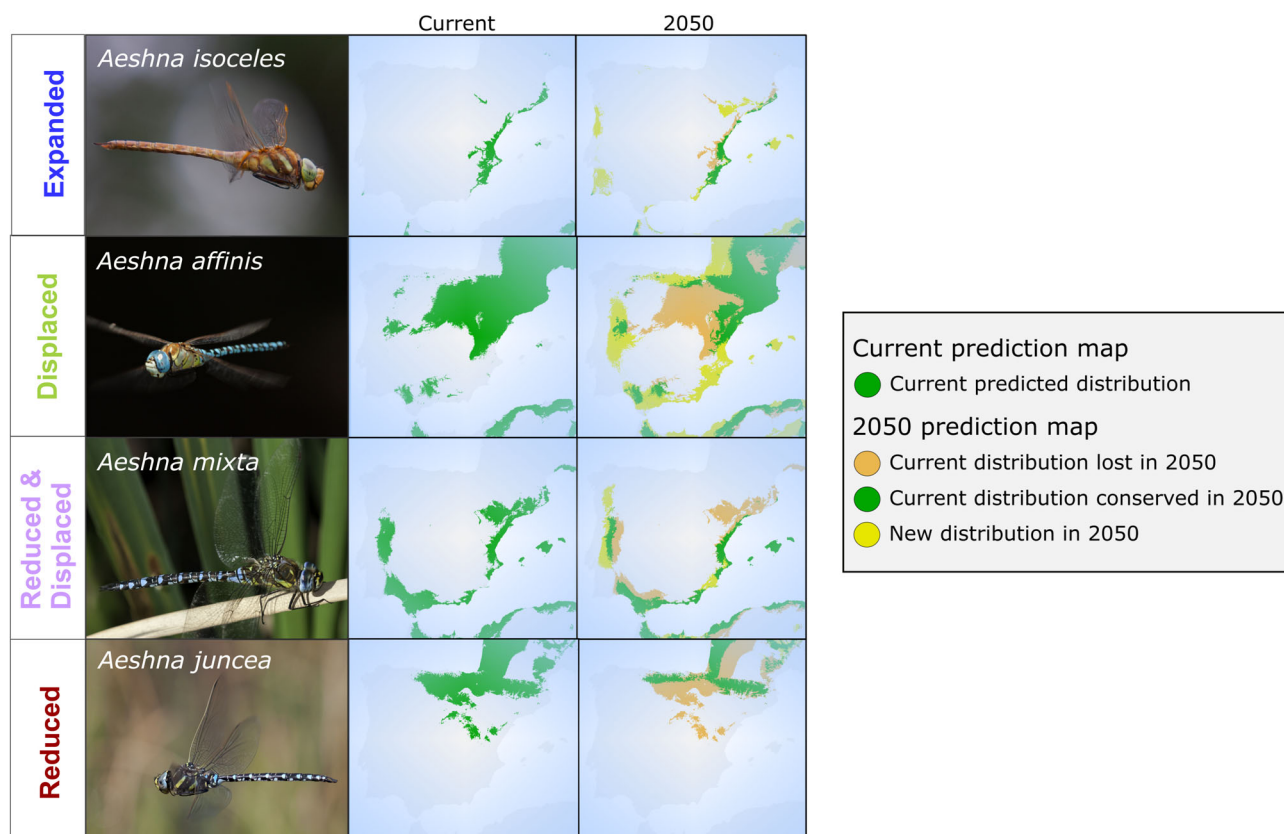


FIGURE 3 Examples of different categories of range-shift responses obtained with the predictions of the RCP 8.5 scenario for four species within the genus *Aeshna*.

All predictions for 2050 under the RCP 2.6 (low emissions) and the RCP 8.5 (high emissions) scenarios were classified into four range-shift categories because no species was classified as 'Unchanged' (Figure 2; see Supporting information S7). As an example, future distribution for four *Aeshna* species was categorised into four different range-shift categories (Figure 3). The most common range-shift categories were 'Expanded' (45.5% of species) and 'Reduced and displaced' (33.3%) under the RCP 2.6 scenario, whereas the majority of predictions under the RCP 8.5 scenario were clustered into the 'Reduced and displaced' (40.9%) and 'Expanded' (39.4%) categories. The categories 'Displaced' (6.1% of species in both scenarios) and 'Reduced' (15.2% and 13.6%, respectively) were less common.

The first fPCA axis of the functional space (Figure 4a; Supporting information S8) explained 20.85% of the total variability, whereas the second and third axes explained respectively 13.62% and 10.94%. The trait categories most correlated to the first axis were semivoltinism (NCO) together with the opposite patterns of behaviour fliers (FBF) versus perchers (FBP), and the water quality oligotrophic (WO) versus eutrophic (WE). The second axis was mostly related with the opposite directions of absence (DN) versus presence (DY) of egg diapause. The trait categories most correlated with axes (>0.6) were selected (Supporting information S9).

The four traits flight season (FS), oviposition habitat (O), range margins (R) and habitat type (H) significantly discriminated among the

range-shift categories (Figure 4c,d). Species categorised as either 'Expanded' or 'Displaced' lived preferentially in eutrophic waters in lowlands and/or in middle-mountain and their habitat preferences were mostly ponds, wetlands and pools. Within the 'Expanded' category, most species did not have egg diapause, showed multivoltinism and the upper limit of their distribution was the Iberian Peninsula (species expanded from North Africa), whereas species within the category 'Displaced' were characterised by univoltinism, their flight season was in summer and the Iberian Peninsula was their central and lower distribution limit (Mediterranean species). Most species in the category 'Reduced and displaced' were widely distributed in the Iberian Peninsula, lived in streams and rivers in lowlands and at middle elevations and were univoltines. Finally, species in the category 'Reduced' lived in oligotrophic waters in mountain lakes, ponds and pools, the Iberian Peninsula was the lower limit of their distribution (species mainly distributed in Central Europe), and showed semivoltinism and a diapause period.

The maximum likelihood phylogenetic tree (Figure 5) included 2 suborders, 9 families and 73 species (80.2% of the entire species list), and the final matrix contained 8649 bp. The species coverage for each gene ranged between 57.53% and 72.60% (Supporting information S5). The two suborders, Anisoptera and Zygoptera were monophyletic (Figure 5), in agreement with previous phylogenetic studies (Dijkstra & Kalkman, 2012; Dumont et al., 2010). Zygoptera

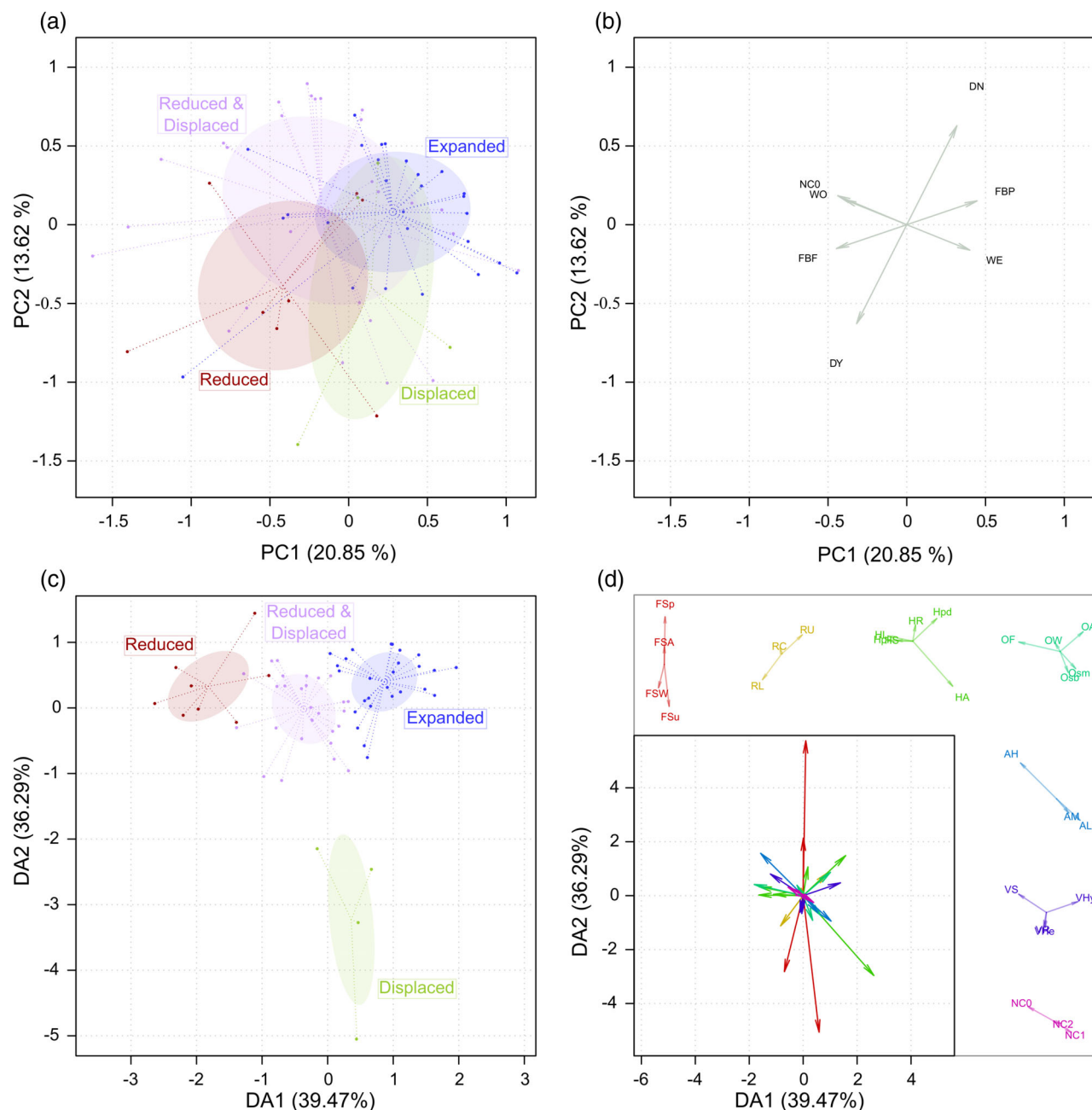


FIGURE 4 Fuzzy principal component analysis (fPCA). (a) Distribution of species on the generated functional space. Ellipses cluster the species belonging to the same category of range-shift responses, each colour and letters corresponding to a different range-shift response as in Figures 1a and 2; (b) arrows represent the most weighted trait categories for principal components 1 and 2 on fPCA (see trait abbreviation in Supporting information S2); (c) distribution of species by discriminant analysis (DA) space; (d) significant traits that discriminated between the categories of range-shift responses in DA. From left to right and from up to down: flight season (FS); range margins (R); habitat type (H); oviposition habitat (O); river zonation (A); vegetation at the breeding site (V); and voltinism (NC). The categories within a trait are depicted in the same colour.

was divided into two supported clades, 'Lestomorphs' (including genera *Lestes* and *Sympecma*, clade 'A', Figure 5) and 'non-Lestomorphs', which included the families Platynemididae, Coenagrionidae and Calopterygidae (clade 'B'). The suborder Anisoptera was separated into two clades ('C' and 'D'). The first clade was well-resolved (Cordulegasteridae, Gomphidae and Aeshnidae), and was coincident with Dumont et al. (2010). However, the monophyletic family

Libellulidae in the clade 'D' did not have all nodes supported, but genera positions agreed with Dijkstra and Kalkman (2012), except for *Zygonyx* and *Crocothemis* that were placed together. In clade 'D', Corduliidae was paraphyletic as in Dijkstra et al. (2013).

Phylogenetic trait conservatism was rejected based on Pagel's λ and Blomberg's K statistics for the species scores on the first and second axes on the fPCA ($\lambda = 0.7250$ and $K = 0.0059$ on the first fPCA

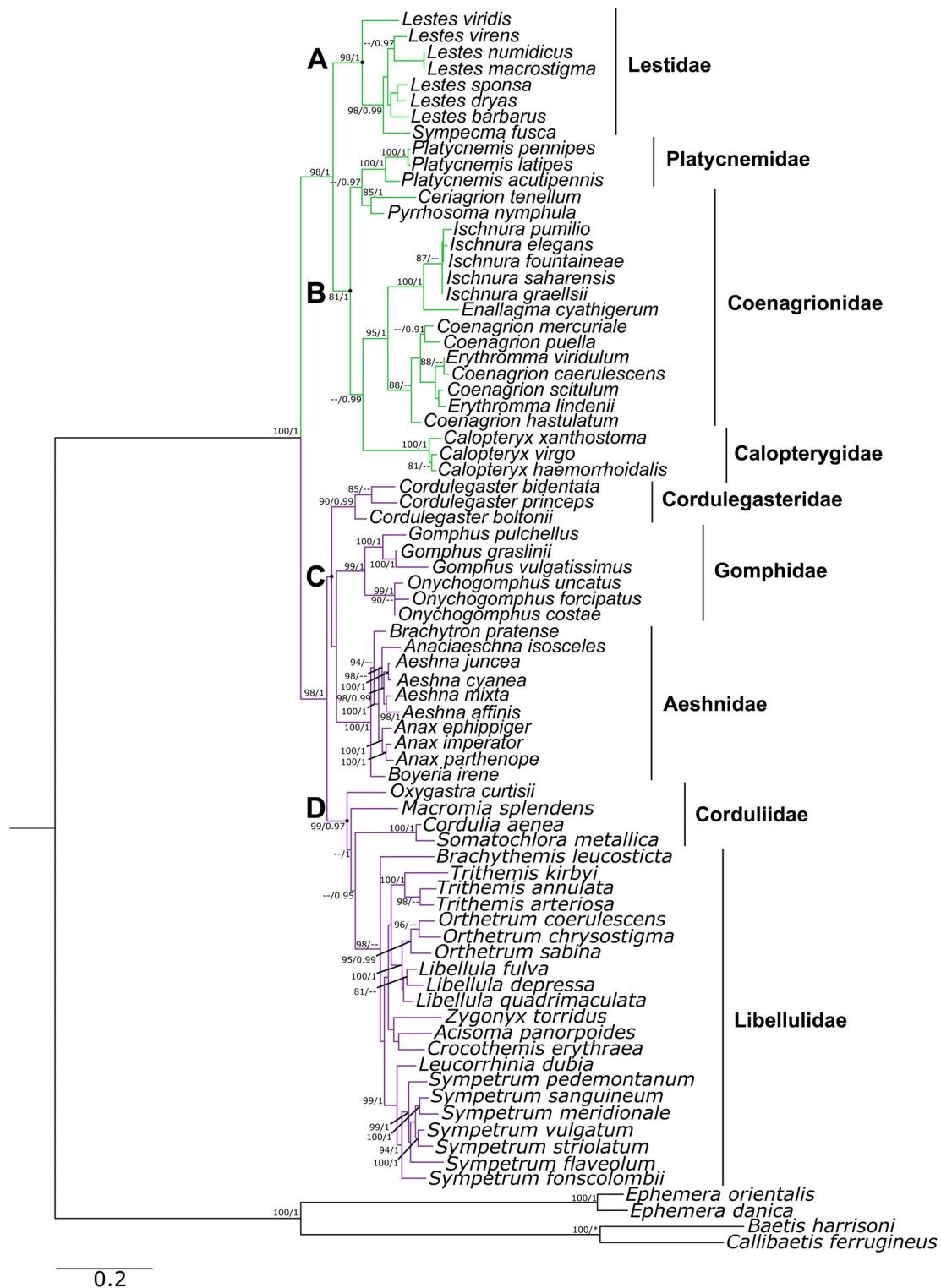


FIGURE 5 Maximum-likelihood phylogenetic tree of the COI, 12S, 16S, 18S and 28S genes including 73 species of Odonates (80.2% of the species list, Supporting information S1 and S5) belonging to 9 families and three outgroup genera of Ephemeroptera. Branch support percentage is indicated: maximum likelihood bootstrap (>80)/Bayesian inference (>0.95). -- refers to clades that appeared in the analysis but with support values below the levels indicated above. Suborder classification is given by coloured branches: Zygoptera (green) and Anisoptera (purple). Four clades are indicated: 'A' for Lestomorphs and 'B' for the non-Lestomorphs clades within Zygoptera. Anisoptera is separated into clades 'C' and 'D'.

axis, and $\lambda = 0.9232$ and $K = 0.0042$ on the second). Moreover, the phylogenetic relatedness was unrelated to the defined range-shift categories for the two climatic scenarios ($\lambda = 0.00$ and $K = 0.1367$, and $\lambda = 0.00$ and $K = 0.1209$, respectively under the RCP 2.6 and the RCP 8.5 scenarios).

DISCUSSION

The modelled future climate in the Iberian Peninsula will be characterised by both warming and increased seasonality, which should likely involve a higher frequency of droughts, as previously reported (Giorgi & Lionello, 2008; Valdes-Abellan et al., 2017). As a consequence, our SDMs anticipated changes in the distribution of suitable habitats for all Odonata species, clustering species into distinct range-shift categories. The 'Expanded' and 'Reduced and displaced' were the two most common responses. Based on the functional space generated, the distinct range-shift categories were most strongly associated to larval development rate, the duration of flight season, permanence of the preferred habitat type and oviposition habitat. Moreover, the three biogeographic origins, North Africa, the Mediterranean basin and Central Europe (Torralba-Burrial & Ocharan Larrondo, 2007), also discriminated among categories. Central European species were mainly clustered in the 'Reduced and displaced' category, whereas the North African and Mediterranean species were categorised as 'Expanded'. All traits studied and the range-shift categories were decoupled from the phylogenetic relatedness between species, which rejects phylogenetic trait conservatism and reveals that the phylogenetic relatedness of odonates cannot anticipate a range-shift category, in contrast with results found for plants (Davis et al., 2010; Fei et al., 2017). Hence, future biodiversity conservation plans for facing climate change of odonates should be based on functional traits rather than on lineage identity, as illustrated by the opposite responses found across the closely related *Aeshna* species (Figure 3).

On the functional space, the anisopterans *Diplacodes lefebvrei*, *Trithemis kirbyi*, *Sympetrum fonscolombii*, *Orthetrum nitidinerve*, *Trithemis annulata*, *Crocothemis erythraea* and *Paragomphus genei* were located at one side of the first fPCA axis and all of them will potentially expand their potential distribution (Supporting information S8). These expanding species were recorded in temporary wetlands and ephemeral streams and can occur also in artificial lakes or reservoirs (Martín et al., 2016). 'Expanded' species prefer oviposition in water without vegetation (exophytic), they show a long flight period covering several seasons and they were also recorded in North Africa, where they are considered widespread desert species (Suhling et al., 2003). The response mechanisms of African odonates such as *C. erythraea*, *T. kirbyi ardens* and *P. genei* for inhabiting temporary wetlands and ephemeral streams in desert and semi-desert areas are to be highly mobile and multivoltine (e.g., short embryonic and larval development) with overlapping annual generations in extreme habitats (Suhling et al., 2003). Moreover, some of these species rest in the 'obelisk position', that is, orientating the tip of the abdomen directly

towards the sun to reduce body exposure to heat as a means of thermoregulation (Corbet, 2004). Overall, our findings reveal the potential mechanistic drivers promoting future expansions associated to climate warming: multivoltinism, short larval development time, early beginning and long flight period, which are traits commonly found in species that inhabit temporary habitats (Hassall & Thompson, 2008). As dryland expansion accelerates under climate change (Huang et al., 2016), the species categorised as 'Expanded' will be favoured, as recently recorded for *T. kirbyi*, which has been establishing populations across the entire Iberian Mediterranean basin (≈ 1000 km long) within the last decade.

Opposed to 'Expanded' species on the first fPCA axis were the species categorised as either 'Reduced and displaced' or 'Reduced', including *Cordulegaster boltonii*, *Aeshna juncea*, *Boyeria irene*, *Cordulegaster princeps*, *Onychogomphus uncatus*, *Macromia splendens* and *Oxygastra curtisii*. Most of these 'Reduced and displaced' or 'Reduced' species are semivoltine and restricted to permanent streams and small rivers, with the exceptions of *M. splendens*, which prefers warm oligotrophic lakes, and *A. juncea*, which inhabits isolated high-elevation lakes in the Pyrenees (Dijkstra, 2006; Martín et al., 2016). The oviposition mode of these species is mainly exophytic (except for *Aeshna* and *Boyeria*, which are endophytic) but more related to gravel than to water, as found for species categorised as 'Expanded'. Moreover, 'Reduced and displaced' or 'Reduced' species showed short flying seasons, often centred in summer, which contrasts with the earlier beginning and prolongation of the flight season that occurs in many insect species favoured by recent global warming (Altermatt, 2009; Dingemanse & Kalkman, 2008; Hassall et al., 2007), as found also for the Iberian 'Expanded' odonates. Consequently, the predicted future decrease in the frequency of permanent habitats can severely impact the populations and distribution of species categorised as 'Reduced and displaced' or 'Reduced'. The mechanistic drivers that may allow these species to reduce the impact of such habitat loss and to acclimate to the new conditions are to increase development speed in order to produce one or more generations per year and to emerge earlier in the season (Lancaster et al., 2016). A good example of phenotypic plasticity in development rates (i.e., voltinism flexibility) favouring a latitudinal adaptation to thermal gradient is *Gomphus vulgatissimus*. This widely distributed European species shows a decrease in voltinism from southern to northern latitudes and earlier emergence as temperatures increase and it is currently expanding its range limits to the north (Braune et al., 2008; Richter et al., 2008). Hence, apart from considering the entire range of distribution (Hassall, 2012), a finer classification of intraspecific functional traits variability across a latitudinal gradient seems to be crucial for improving the precision when anticipating future distribution shifts and migrations into new regions.

Another biological trait that highly contributed to splitting species into different range-shift categories was the absence versus presence of egg diapause. This trait is directly related to voltinism, even though diapause was not selected as a statistically significant trait for discriminating among categories. Diapause induction, or development interruption, has been favoured to cope with seasonal fluctuations as it

allows species to avoid unfavourable environmental conditions such as cold (hibernation) or drought (aestivation) by means of resistant forms of either eggs, larvae or adults in species that breed in temporary waters (Suhling et al., 2015). Photoperiod and temperature have been suggested as the mechanisms controlling diapause. For instance, winter diapause is thought to be crucial for the success of odonates in colonising temperate regions and subarctic habitats from their tropical areas of origin (Pritchard, 2008). However, we found that Eurosiberian and Holarctic species (*sensu stricto* in Torralba-Burrial & Ocharan Larrondo, 2007) with diapause in the egg stage such as *Aeshna mixta*, *Sympetrum striolatum*, *Lestes sponsa* and *Lestes dryas* were categorised as either 'Reduced and displaced' or 'Reduced' in the Iberian Peninsula, but are found to moderately increase their range in Sweden (Termaat et al., 2019). Therefore, our predictions suggest future climate warming in the Iberian Peninsula will not favour this strategy and their southernmost limit of distribution should move northward, as previously suggested (Hassall & Thompson, 2008).

Previous studies revealed clear differences in functional trait composition across freshwater macroinvertebrate lineages (i.e. Odonata, Ephemeroptera or Trichoptera) due to evolutionary constraints on ancestral ecological characteristics that limited trait disparification (Múrria et al., 2018; Poff et al., 2006). Until now the few studies that have tested phylogenetic trait conservatism within an individual lineage of freshwater insects supported phylogenetic conservatism (García-Raventós et al., 2021; Múrria et al., 2012; Pauls et al., 2008). In general, speciation and trait disparification are mainly driven either by environmental filtering if species retain similar functionalities, or by interspecific competition, which promotes trait divergence across species (Webb et al., 2002). For Odonates, the rejection of phylogenetic trait conservatism suggests the importance of interspecific competition and indicates phylogenetically related species acquired distinct functional traits to occupy different habitats or climates. Given that a particular trait configuration was assigned to one range-shift category independently of species identity, as found among closely related *Aeshna* species (Figure 3), phylogenetic relatedness of odonates is a weak factor for anticipating species responses to climate change.

Overall, our modelling approach is consistent with previous long-term empirical studies that revealed the considerable effects of future climate warming on range-shift responses of odonates (Angert et al., 2011; Hassall & Thompson, 2008; Hickling et al., 2005). Moreover, other forms of global change such as water extraction or fertiliser and human waste inputs that increase eutrophication may act synergistically with climate warming to affect biodiversity. Therefore, other traits such as tolerance to salinity, which is not considered here, should be added in future studies for a complete understanding of the effects of global change on freshwater biodiversity (Albert et al., 2021; Jeppesen et al., 2020).

If we consider odonates as model organisms, similar responses could also be expected for other freshwater macroinvertebrates. Hence, species with long life-cycles and aerial adults with a limited flight period in summer or autumn inhabiting oligotrophic permanent waters or high- and mid-elevation habitats should reduce and/or displace their current ranges. In contrast, species able to tolerate salinity,

inhabiting eutrophic and temporary freshwater habitats, with fast development rates and lack of diapause should expand their range. Importantly, understanding trait plasticity to adapt locally in critical traits associated to climate warming such as larval growth rates, timings of adult emergence and flight period (e.g., Maynou & Martín, 2019), should be crucial for anticipating the vulnerability of species categorised as 'Reduced and displaced' and 'Reduced'.

AUTHOR CONTRIBUTIONS

Aida Viza: Conceptualization (equal); data curation (lead); formal analysis (lead); investigation (equal); methodology (equal); writing – original draft (lead); writing – review and editing (lead). **Aina García-Raventós:** Conceptualization (equal); data curation (supporting); formal analysis (supporting); investigation (equal); methodology (equal); supervision (supporting); writing – original draft (supporting); writing – review and editing (supporting). **Joan Lluís Riera:** Formal analysis (supporting); methodology (equal); supervision (supporting); validation (equal); writing – original draft (supporting); writing – review and editing (supporting). **Xavier Maynou:** Data curation (supporting); investigation (supporting); resources (lead); supervision (supporting); writing – original draft (supporting); writing – review and editing (supporting). **Ricard Martín:** Data curation (supporting); investigation (supporting); resources (lead); supervision (supporting); writing – original draft (supporting); writing – review and editing (supporting). **Florent Prunier:** Resources (lead); writing – original draft (supporting). **Mohamed El Haissoufi:** Resources (supporting). **Cesc Murria:** Conceptualization (equal); data curation (supporting); formal analysis (supporting); investigation (equal); methodology (equal); supervision (lead); validation (equal); writing – original draft (supporting); writing – review and editing (supporting).

ACKNOWLEDGEMENTS

We would like to thank Núria Bonada and Tony Herrera for their feedback during the initial steps of the project that greatly improved the study design, and the members of the research group FEHM (Freshwater Ecology, Hydrology and Management) for enriching discussions. Also, we would like to thank all members of naturalist groups that shared their data with us. Cesc Múrria was supported by a Beatriu de Pinós postdoctoral fellowship (BP-DGR-2011) from Agència de Gestió d'Ajuts Universitaris i de Recerca, Generalitat de Catalunya.

CONFLICT OF INTEREST STATEMENT

All authors declare no competing interests.

DATA AVAILABILITY STATEMENT

Public data of six datasets used are: (1) Oxygastra data, which can be accessed on gbif.org at <https://www.gbif.org/publisher/fa023734-40bb-484a-b6c4-c822ad0d89ad>; (2) ROLA data, available on their website at https://aeaelbosqueanimado.org/project_category/boletin-rola/; (3) the Arthropod Collection of the University of Oviedo, which can be found on their server at <http://digibuo.uniovi.es/dspace/handle/10651/17721>; (4) data from the Valencian Natural History Museum, which was compiled by the Biodiversity Databank of the Valencian Community and can be accessed on gbif.org at <https://www.gbif.org/>

[publisher/ebda1cab-dda8-418b-8f25-df1e71ff5bc2](https://doi.org/10.1016/j.incon.2023.100000); and (5) ANSE data, which is available on their website at <https://www.asociacionanse.org/publicaciones/>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Supporting information S1. Species records used to each distribution model.

Supporting information S2. Model performances by each species. Columns of predictor variables showed variables selected and the corresponded distribution for each model. Distribution codes for GLM: $x^2 + x =$: quadratic regression; $x =$: linear regression. In BRT, variables selected were indicated with a ‘x’.

Supporting information S3. Codes and abbreviations of traits and categories selected.

Supporting information S4. Matrix of traits and their categories for each species.

Supporting information S5. GenBank accession numbers for used sequences of Odonata and Ephemeroptera outgroups for each of the five genes.

Supporting information S6. Phylogenetic tree procedure.

Supporting information S7. Species classification in the proposed categories of range shift and their coordinates in the two different scenarios. Categories: Exp = Expanded; Dis = Displaced; NC = Unchanged; R&D = Reduced & Displaced; Red = Reduced.

Supporting information S8. Species coordinates of functional space axes.

Supporting information S9. Correlation of trait categories with functional space axes.

How to cite this article: Viza, A., Garcia-Raventós, A., Ll. Riera, J., Maynou, X., Martín, R., Prunier, F. et al. (2023) Species-specific functional traits rather than phylogenetic relatedness better predict future range-shift responses of odonates. *Insect Conservation and Diversity*, 16(5), 574–587. Available from: <https://doi.org/10.1111/icad.12647>